

금융투자
상품

2025-1

파생금융상품 — Futures (선물계약)

Futures and Forwards

- **Forwards** (선도계약) – a deferred delivery of a **customized** contract with the sales price agreed on now.
 - , FX() ...
 -> (OTC)
- **Futures** (선물계약) - similar to forward but feature formalized and **standardized** contracts.
- Key difference in futures
 - Standardized contracts create **liquidity**
 - **Marked to market** () -> 0
 - Exchange mitigates **credit risk**
 0 -> exchange - traded()

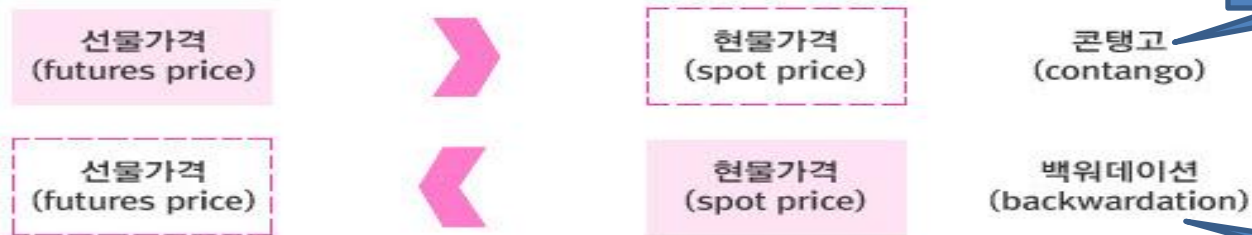
FUTURES	FORWARDS
Traded on exchange	Privately negotiated
Standardized, having an exchange-specified contract unit, expiration, tick size, and notional value	Customized
No counterparty risk, since payment is guaranteed by the exchange clearing house	Credit default risk, since it is privately negotiated, and fully dependent on the counterparty for payment
Actively traded	Non-transferrable
Regulated	Not regulated

Basics of Futures Contracts

- A futures contract is the **obligation** to make or take delivery of the underlying asset at a predetermined price.
- Futures price – the price for the underlying asset is determined today, but settlement is on a future date.
- The futures contract specifies the quantity and quality of the underlying asset and how it will be delivered.

베이스스 (basis)

$$\text{베이스스 (basis)} = \text{선물가격 (futures price)} - \text{현물가격 (spot price)}$$



정상시장

역조시장

* 이론 선물가격 = 현물가격 + 보유비용(이자비용, 보관비용 등)

선물과 현물을 서로 반대방향으로 동시에 투자해 차익을 노리는데 이를 '차익매매(arbitrage)'

매수차익거래
(베이스스 ↑)

선물가격
고평가



선물매도
현물매수

매도차익거래
(베이스스 ↓)

선물가격
저평가



선물매수
현물매도

균형가격 도달
(베이스스=0)

"정상적인 시장에서의 선물이론가격"

위의 식에서 확인할 수 있듯이 금리가 높고 만기일까지 잔존기간이 길수록 선물이론가격은 높게 나타납니다. 따라서 정상적인 시장조건에서는 선물시장가격은 현물가격보다 비싼 값에 거래가 됩니다.

"하락장에서의 선물이론가격"

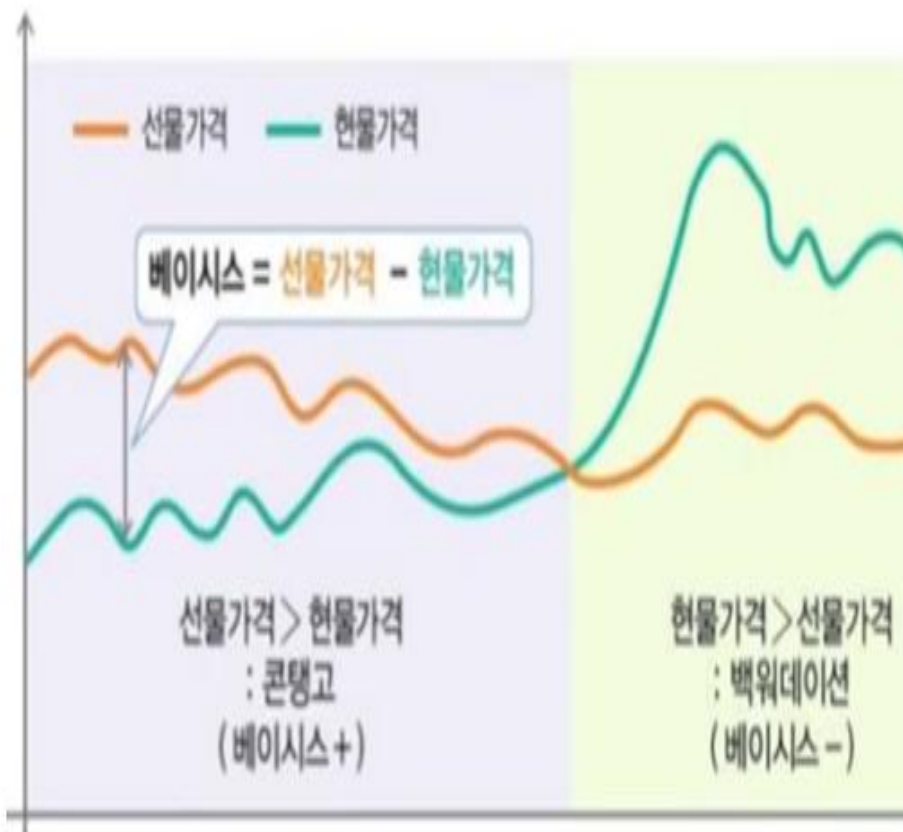
하지만 경기침체나 주가에 악영향을 주는 악재가 있을 시에는 현물가격이 하락하게 됩니다. 이런 경우에는 단기 금리는 내려갈 것이고 배당수익률은 높아지게 됩니다. 그러면 위에 식에서

"현물가격 x (단기금리 - 배당수익률) x 잔존기간"

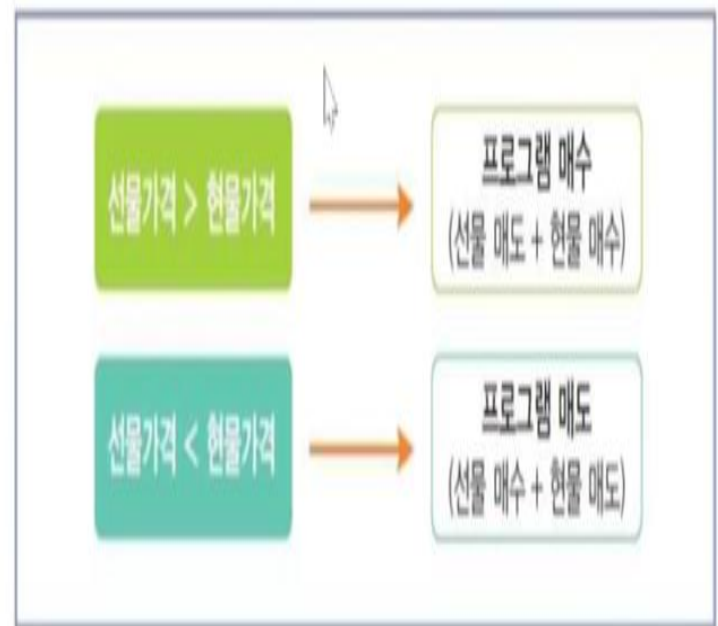
항목이 "-"(마이너스)로 바뀌기 때문에 선물이론가격은 현재 현물가격에 비해 낮게 형성됩니다.

"만기일에 가까운 선물이론가격"

만기일이 될수록 선물이론가격은 현물가격에 접근하게 됩니다. 위의 선물이론가격 공식에서 에서 보면 만기일이 다가올수록 이자를 부담하는 기간이 짧아지므로 금리의 영향은 점점 작아집니다. 그러다가 만기일에는 금리의 영향이 "0" (제로)가 됩니다. 그래서 만기일이 되면 선물이론가격과 현물가격은 같아지게 됩니다.



[프로그램 매매]



Basics of Futures Contracts

- **Position:**

Long – a commitment to purchase the commodity on the delivery date.

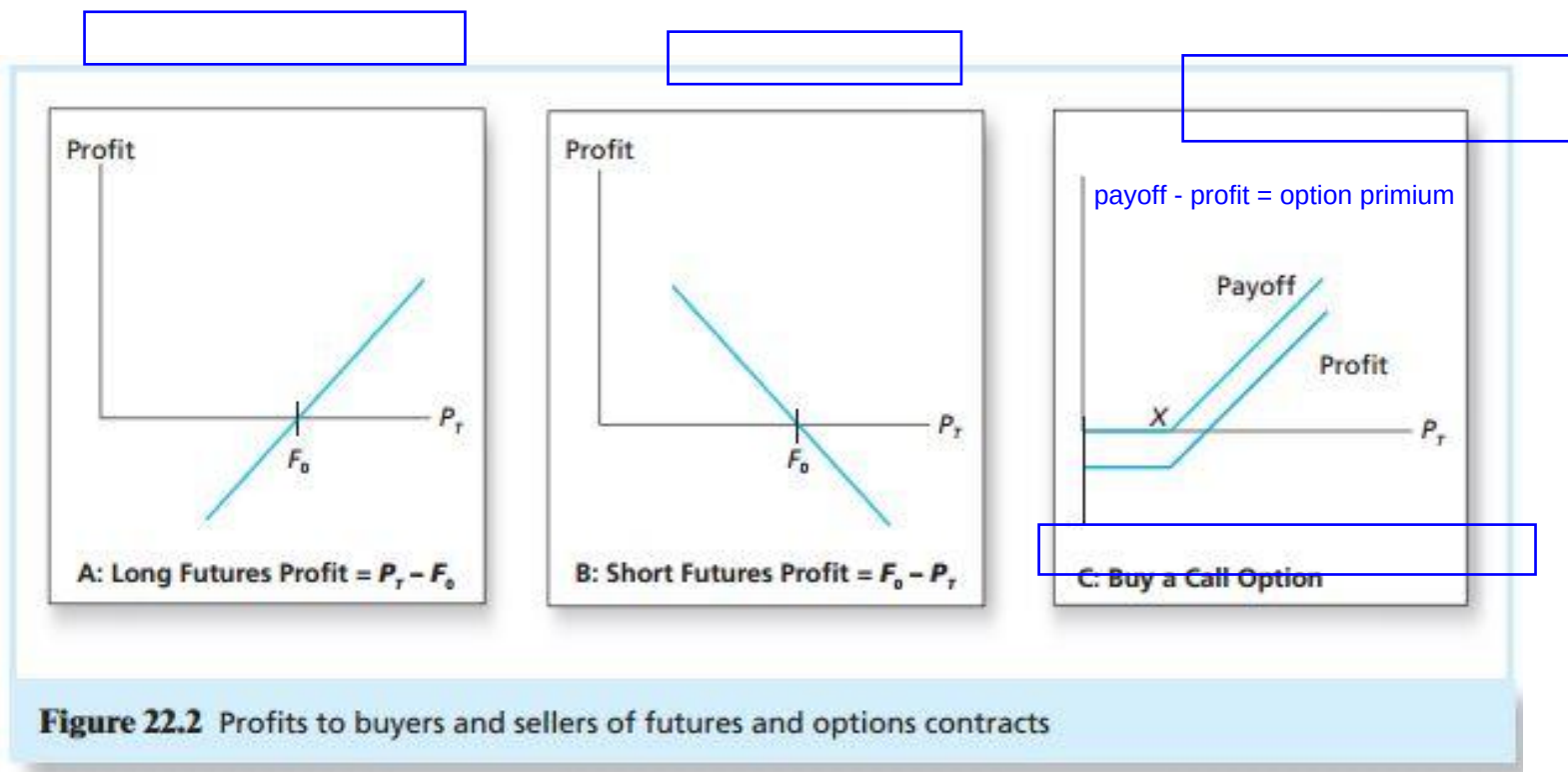
Short – a commitment to sell the commodity on the delivery date.

- Futures are traded on margin.
- At the time the contract is entered into, no money changes hands.

Basics of Futures Contracts

- Profit to long = $\text{Spot price at maturity} - \text{Original futures price}$
- Profit to short = $\text{Original futures price} - \text{Spot price at maturity}$
- The futures contract is a **zero-sum game**, which means gains and losses net out to zero.

Profits to Buyers and Sellers of Futures and Option Contracts



- Profit is zero when the ultimate spot price, P_T equals the initial futures price, F_0 .
- Unlike a call option, the payoff to the long position can be negative because the futures trader cannot walk away from the contract if it is not profitable.

기초자산 유형

- Futures contracts are traded on a wide variety of assets in four main categories:

1. Agricultural commodities

2. Metals and minerals

3. Foreign currencies

4. Financial assets

Trading Mechanics

- Electronic trading has mostly displaced floor trading.
- CBOT and CME merged in 2007 to form CME Group.
- The exchange acts as a clearing house and counterparty to both sides of the trade.
- The net position of the clearing house is zero.

Trading Mechanics

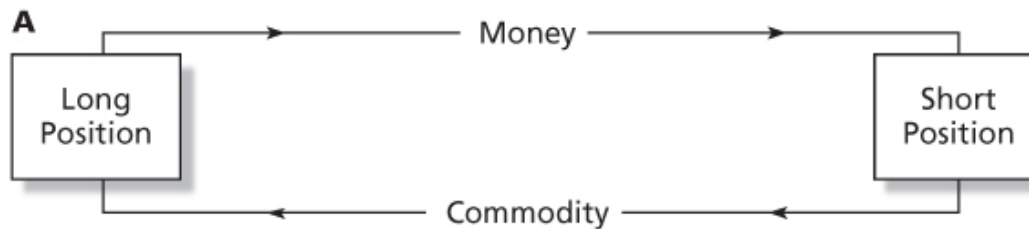
open interest : (,; .) -> speculation()

- Open interest is the number of contracts outstanding.
- If you are currently long, you simply instruct your broker to enter the short side of a contract to close out your position.
- Most futures contracts are closed out by reversing trades.

; 0 (), ? -> ., OK
- Only 1-3% of contracts result in actual delivery of the underlying commodity.

Trading without a Clearinghouse; Trading with a Clearinghouse

(forward)



(future)

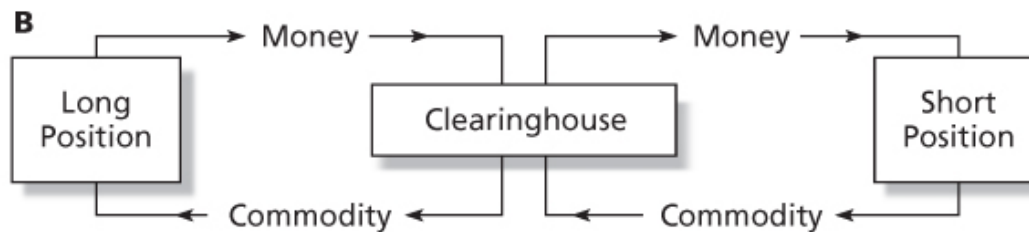


Figure 22.3 *Panel A*, Trading without a clearinghouse. *Panel B*, Trading with a clearinghouse.

Margin and Marking to Market

- Marking to Market - each day the profits or losses from the new futures price are paid over or subtracted from the account
- Convergence of Price - as maturity approaches the spot and futures price converge

Margin and Trading Arrangements

- Initial Margin - funds or interest-earning securities deposited to provide capital to absorb losses
- Maintenance margin – level at which the account must be replenished or position reduced
- Margin call - when the maintenance margin is reached, broker will ask for additional margin funds

Trading Strategies



Speculators

- seek to profit from price movement
 - Short: believe price will fall
 - long : believe price will rise



Hedgers

- seek protection from price movement
 - long hedge: protecting against a rise in purchase price
 - short hedge: protecting against a fall in selling price

Basis and Basis Risk

- Before maturity, F_T may differ substantially from the current spot price.
-  Basis Risk - variability in the basis means that gains and losses on the contract and the asset may not perfectly offset if liquidated before maturity.

Futures Pricing

Spot-futures parity theorem - two ways to acquire an asset for some date in the future:

1) 2) $\rightarrow$ 1),2) ; 1),2) ; = parity

1. Purchase it now and store it
 2. Take a long position in futures
- These two strategies must have the same market determined costs

Spot-Futures Parity Theorem

- With a perfect hedge, the futures payoff is certain: no risk
- A perfect hedge should earn the riskless rate of return.
- This relationship can be used to develop the futures pricing relationship.

Hedge Example

- Investor holds \$1000 in a mutual fund indexed to the S&P 500.
- Assume dividends of \$20 will be paid on the index fund at the end of the year.
- A futures contract with delivery in one year is available for \$1,010.
- The investor hedges by selling or shorting one contract .

Hedge Example Outcomes

	990	1,010	1,030
Value of S_T	990	1,010	1,030
Payoff on Short (1,010 - S_T)	20	0	-20
Dividend Income 20	<u>20</u>	<u>20</u>	<u>20</u>
Total	1,030	1,030	1,030
	1,030		

Rate of Return for the Hedge

$$\frac{(F_0 + D) - S_0}{S_0} = \frac{(1,010 + 20) - 1,000}{1,000} = 3\%$$

The Spot-Futures Parity Theorem

$$\frac{\overset{\text{future}}{(F_0 + \boxed{D})} - \overset{\text{spot}}{S_0}}{S_0} = r_f \boxed{}$$

Rearranging terms

$$\boxed{} F_0 = S_0 (1 + r_f) - D = S_0 (1 + r_f - \boxed{d})$$

$$d = \frac{D}{S_0}$$

Arbitrage Possibilities

- If spot-futures parity is not observed, then arbitrage is possible.
- If the futures price is too high, short the futures and acquire the stock by borrowing the money at the risk free rate.
- If the futures price is too low, go long futures, short the stock and invest the proceeds at the risk free rate.

Spread Pricing: Parity for Spreads

$$F(T_1) = S_0 (1 + r_f - d)^{T_1}$$

$$F(T_2) = S_0 (1 + r_f - d)^{T_2}$$

$$F(T_2) = F(T_1)(1 + r_f - d)^{(T_2 - T_1)}$$

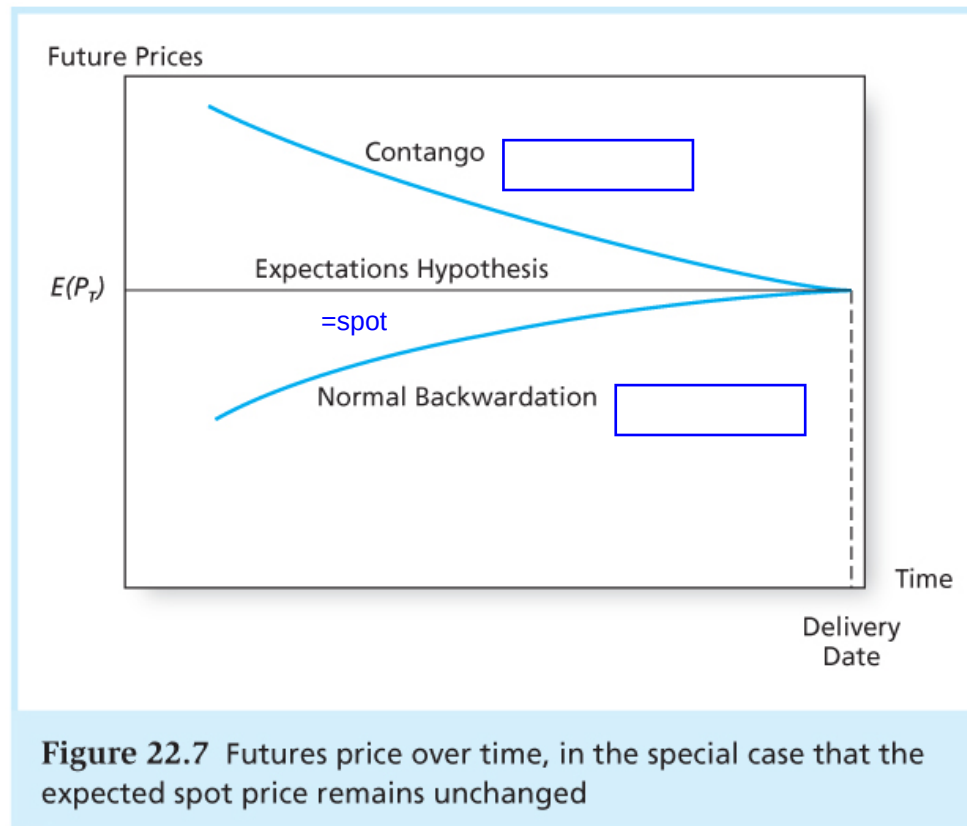
Spreads

- If the risk-free rate is greater than the dividend yield ($r_f > d$), then the futures price will be higher on longer maturity contracts.
- If $r_f < d$, longer maturity futures prices will be lower.
- For futures contracts on commodities that pay no dividend, $d=0$, F must increase as time to maturity increases.

Futures Prices vs. Expected Spot Prices

- Expectations
- Normal Backwardation
- Contango
- Modern Portfolio Theory

Futures Price Over Time, Special Case



Futures

- Futures can be used to hedge specific sources of risk.
- Hedging instruments include:
 - Foreign exchange futures
 - Stock index futures
 - Interest rate futures
 - Swaps
 - Commodity futures

Pricing on Foreign Exchange Futures

- Interest rate parity theorem

Developed using the US Dollar and

British Pound

where

$$F_0 = E_0 \left(\frac{1 + r_{US}}{1 + r_{UK}} \right)^T$$

law of one price in financial market-> ()

F_0 is today's forward rate

E_0 is the current spot rate

Text Pricing Example

(or)

$$r_{us} = 4\%$$

$$r_{uk} = 5\%$$

$$E_0 = \$2.00 \text{ per pound}$$

$$T = 1 \text{ yr}$$

$$F_0 = \$2.00 \left(\frac{1.04}{1.05} \right)^1 = \$1.981$$

1 ? -> (1.04/1.05)^1

Hedging Foreign Exchange Risk

A US exporter wants to protect against a decline in profit that would result from depreciation of the pound. The current futures price is \$2/£1.

Suppose $F_T = \$1.90$?

- The exporter anticipates a profit loss of \$200,000 if the pound declines by \$.10
- Short or sell pounds for future delivery to avoid the exposure.

$$1 = 1.9??$$

$$0.1 = 200() * 0.1(?) = 20()$$

;;

Hedge Ratio for Foreign Exchange Example

$$H = \frac{\$200,000 \text{ per } \$.10 \text{ change in } \frac{\$}{\pounds} \text{ exchange rate}}{\$.10 \text{ per pound delivered per } \$.10 \text{ change in } \frac{\$}{\pounds} \text{ exchange rate}}$$

Hedge Ratio in contracts

Each contract is for 62,500 pounds or \$6,250 per a \$.10 change

$\$200,000 / \$6,250 = 32$ contracts

$/ : 2/1 \rightarrow 1.9/1 \rightarrow 62500 * (2-1.9) / = 6$

62500,

Stock Index Futures Contracts

- Available on both domestic and international stocks
- Settled in cash
- Advantages over direct stock purchase
 - lower transaction costs
 - better for timing or allocation strategies
 - takes less time to acquire the portfolio

Major Stock-Index Futures

Contract	Underlying Market Index	Contract Size	Exchange
S&P 500	Standard & Poor's 500 index. A value-weighted arithmetic average of 500 stocks.	\$250 times S&P 500 index	Chicago Mercantile Exchange
Dow Jones Industrial Average (DJIA)	Dow Jones Industrial Average. Price-weighted average of 30 firms.	\$10 times index	Chicago Board of Trade
Russell 2000	Index of 2,000 smaller firms.	\$100 times index	Intercontinental Exchange (ICE)
NASDAQ 100	Value-weighted arithmetic average of 100 of the largest over-the-counter stocks.	\$100 times index	Chicago Mercantile Exchange
Nikkei 225	Nikkei 225 stock average.	\$5 times Nikkei Index	Chicago Mercantile Exchange
FTSE 100	Financial Times Stock Exchange Index of 100 U.K. firms.	£10 times FTSE Index	London International Financial Futures Exchange
DAX-30	Index of 30 German stocks.	25 euros times index	Eurex
CAC-40	Index of 40 French stocks.	10 euros times index	Euronext Paris
Hang Seng	Value-weighted index of largest firms in Hong Kong.	50 Hong Kong dollars times index	Hong Kong Exchange

Table 23.1

Sample of stock-index futures

Creating Synthetic Positions with Futures

- Index futures let investors participate in broad market movements without actually buying or selling large amounts of stock.
- Results:
 - Cheaper and more flexible
 - Synthetic position; instead of holding or shorting all of the actual stocks in the index, you are long or short the index futures

Creating Synthetic Positions with Futures

- Speculators on broad market moves are major players in the index futures market.
 - Strategy: Buy and hold T-bills and vary the position in market-index futures contracts.
 - If bullish, then long futures
 - If bearish, then short futures

Hedging Systematic Risk

To protect against a decline in stock prices, short the appropriate number of futures index contracts.

- Less costly and quicker
- Use the beta for the portfolio to determine the hedge ratio.

$i \rightarrow i_0$

Hedging Systematic Risk Example

Portfolio Beta = .8

S&P 500 = 1,000

Decrease = 2.5%

S&P falls to 975

Portfolio Value = \$30 million

Projected loss if market declines by 2.5% = $(.8)(2.5\%) = 2\%$

2% of \$30 million = \$600,000

Each S&P500 index contract will change \$6,250 for a 2.5% change in the index. (The contract multiplier is \$250).

$$\$6250 = 1000(= \text{S\&P 500}) * \$250(=) * 0.025(=2.5\%)$$

Hedge Ratio Example

$$\begin{aligned} H &= \frac{\text{Change in the portfolio value}}{\text{Profit on one futures contract}} \\ &= \frac{\$600,000}{\$6,250} = 96 \text{ contracts short} \end{aligned}$$

Uses of Interest Rate Hedges

- A bond fund manager may seek to protect gains against a rise in rates.
- Corporations planning to issue debt securities want to protect against a rise in rates.
- A pension fund with large cash inflows may hedge against a decline in rates for a planned future investment.

Hedging Interest Rate Risk Example

Portfolio value = \$10 million

Modified duration = 9 years

If rates rise by 10 basis points (.1%), then

Change in value = (9) (.1%) = .9% or \$90,000

Price value of a basis point (PVBPP) = \$90,000 / 10 =
\$9,000 per basis point

Hedge Ratio Example

$$H = \frac{\text{PVBP for the portfolio}}{\text{PVBP for the hedge vehicle}}$$

$$= \frac{\$9,000}{\$90} = 100 \text{ T-Bond contracts}$$

(T-bond 10y) -> = 1000

90(=) -> 89.1 (0.9, 0.1%) 89.1 = 1000 * 0.9 =

hedge -> lock-in()

Commodity Futures Pricing



- General principles that apply to stocks apply to commodities. However...
- Carrying costs are more for commodities.
- Spoilage may be a concern.



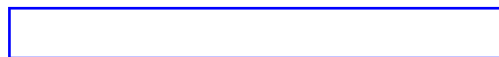
Commodity Futures Pricing

Let F_0 = futures price, P_0 = cash price of the asset, and **C = Carrying cost**

$$F_0 = P_0(1 + r_f) + C$$

Now define $c = C / P_0$ then :

$$F_0 = P_0(1 + r_f + c)$$



Commodity Futures Pricing Example

- The T-bill rate is 5%, the market risk premium is 8%, and the beta for orange juice is 0.117. $\rightarrow 0.117(0.08)$
- Orange juice discount rate is $5\% + .117(8\%) = 5.94\%$.
- Let the expected spot price in 6 months be \$1.45.

$$\begin{aligned} & \$1.45 / (1.0594)^{0.5} \\ & = \$1.409 = \text{PV juice} \end{aligned}$$

commodity : carrying cost (/ ..) financial asset

$$\begin{aligned} & F_0 / (1.05)^{0.5} \\ & = 0.976 F_0 = \text{PV futures} \end{aligned}$$

$$\begin{aligned} 0.976 F_0 &= \$1.409 \\ F_0 &= \$1.444 \end{aligned}$$

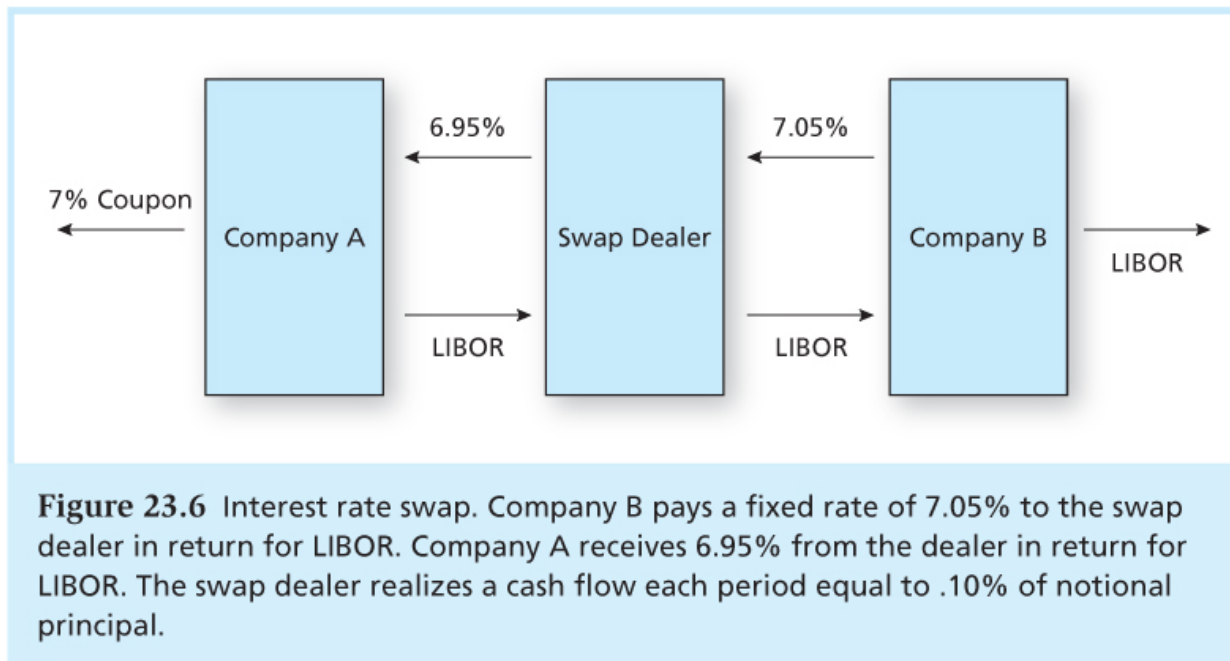
Swaps

- Swaps are multi-period extensions of forward contracts.
- Credit risk on swaps
- An interest rate swap calls for exchanging cash flows based on a fixed rate for cash flows based on a floating rate.
- The foreign exchange swap calls for an exchange of currencies on several future dates.

The Swap Dealer

- Dealer enters a swap with Company A
 - Pays fixed rate and receives LIBOR
- Dealer enters another swap with Company B
 - Pays LIBOR and receives a fixed rate
- When two swaps are combined, dealer's position is effectively neutral on interest rates.

Interest Rate Swap



거래구조



Credit Default Swaps

- Payment on a CDS is tied to the financial status of one or more reference firms.
- Allows two counterparties to take positions on the credit risk of those firms.
- Indexes of CDS have now been introduced.

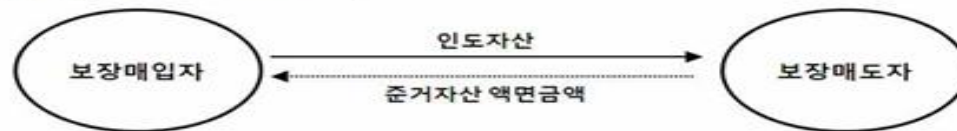
CDS 거래구조



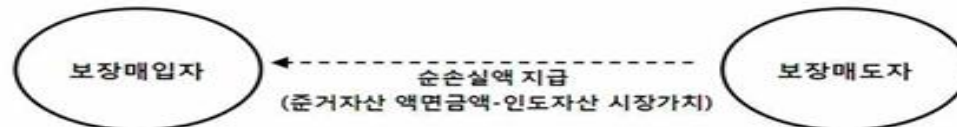
자료: 김인규·오종민(2008), 「신용파생상품시장의 현황과 과제」, 한국은행.

〈그림 1〉 신용부도스왑의 정산방식: 현물결제·현금결제

• 현물결제(Physical Settlement)

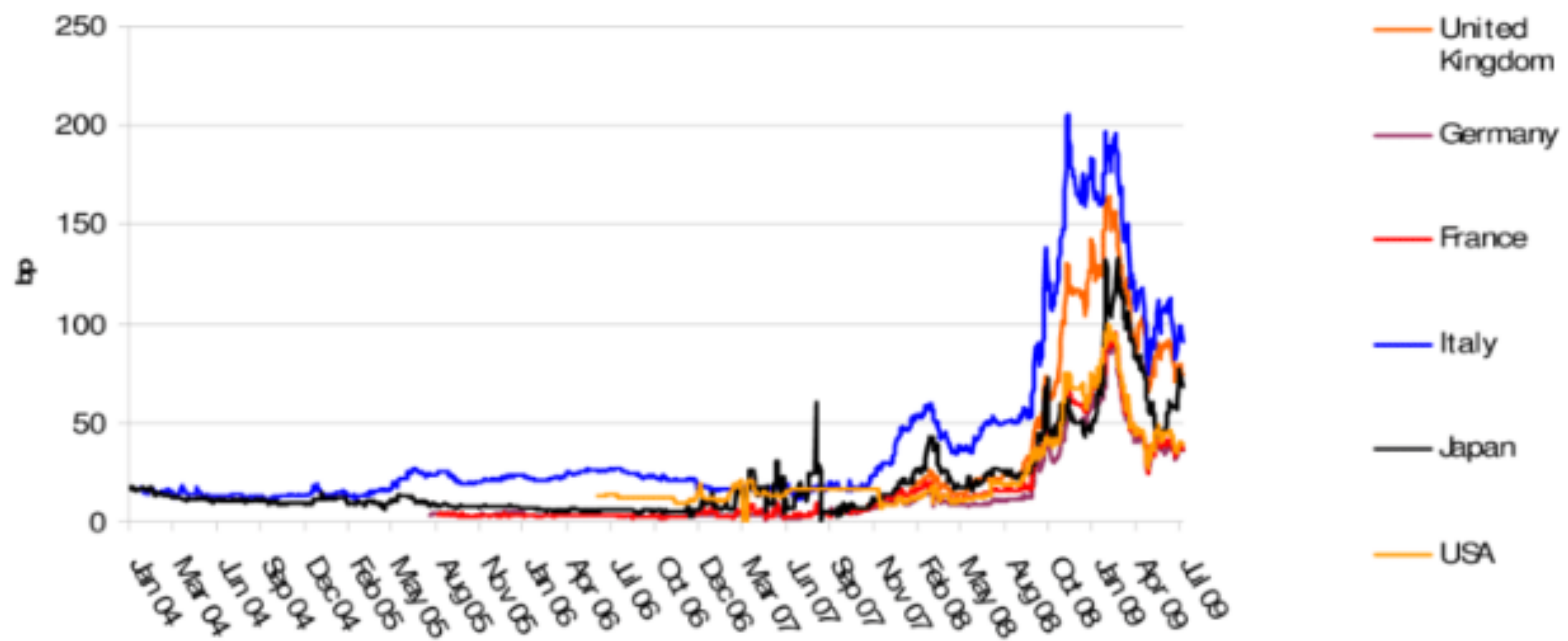


• 현금결제(Cash Settlement)



자료: 김경무·양정용·정연수(2007), 『신용파생상품의 구조와 리스크 분석』.

Sovereigns



Source: Datastream

CDS rates

☐	종목명 ⇅	증가	변동 ⇅	변동 % ⇅	1개월간 ⇅	1년 ⇅	시간
☐	 터키 CDS 5년물	269.78	-5.14	-1.87%	-10.11%	-51.14%	09/05
☐	 이탈리아 CDS 5년물	61.83	-0.04	-0.06%	-6.08%	-45.21%	10/05
☐	 미국 CDS 5년물 USD	36.86	-0.01	-0.03%	+5.13%	-43.55%	10/05
☐	 영국 CDS 5년물	26.31	+0.11	+0.42%	-6.90%	-0.01%	09/05
☐	 스페인 CDS 5년물	35.86	+0.11	+0.31%	-3.91%	-34.08%	09/05
☐	 브라질 CDS 5년물	138.42	-0.05	-0.04%	-6.64%	-37.65%	09/05
☐	 호주 CDS 5년물	13.15	+0.04	+0.31%	-20.06%	-50.94%	10/05
☐	 캐나다 CDS 5년물	39.60	0.00	0.00%	0.00%	+0.40%	21/11
☐	 멕시코 CDS 5년물	93.07	+1.00	+1.09%	-4.08%	-22.93%	09/05
☐	 프랑스 CDS 5년물	25.00	0.00	0.00%	+2.04%	-9.06%	09/05
☐	 이스라엘 CDS 5년물	127.80	-0.01	-0.01%	-3.00%	+99.09%	09/05
☐	 이집트 CDS 5년물	613.86	+0.40	+0.07%	+5.67%	-66.24%	09/05
☐	 중국 CDS 5년물	64.74	+0.01	+0.02%	-8.46%	-14.50%	10/05
☐	 인도네시아 CDS 5년물	72.23	0.00	0.00%	-3.99%	-25.95%	09/05
☐	 인도 CDS 5년물	84.11	+0.01	+0.01%	-0.01%	+2.90%	10/05
☐	 사우디아라비아 CDS 5년물	50.23	-0.04	-0.08%	-13.68%	-23.53%	10/05
☐	 일본 CDS 5년물	24.18	0.00	0.00%	+19.23%	-0.42%	09/05
☐	 스위스 CDS 5년물	5.52	+0.03	+0.55%	-8.61%	-73.13%	09/05
☐	 한국 CDS 5년물	35.09	+0.02	+0.06%	-8.05%	-22.34%	09/05
☐	 남아프리카 공화국 CDS 5년물	222.19	-4.14	-1.83%	-12.31%	-21.67%	09/05
☐	 독일 CDS 5년물	10.01	0.00	0.00%	-4.85%	-28.66%	09/05

<https://kr.investing.com/rates-bonds/world-cds>